

ENTERPRISE SOLUTION REPORT

SafetyOS

An AI-Powered Industrial Safety Intelligence Platform for Zero-Harm Operations across Refining, Chemicals, Steel, Mining and Heavy Manufacturing.

AUTHORING TEAM	CODENAME	MODULES	MICROSERVICES
Team Drona	Halo	27 · 466 Features	48 Services
CLASSIFICATION	TARGET STACK	STANDARDS	REPOSITORY
Confidential — Product Blueprint	Next.js 15 · Go · Python · Neo4j · TimescaleDB	ISA-101 · ISA-18.2 · OSHA · OISD · EU AI Act	github.com/Shubham2786/SafetyOS

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Executive Summary

SafetyOS is an AI-native intelligence layer for industrial operations. It fuses computer vision, OT/SCADA telemetry, IoT wearables, and enterprise workflows into a single explainable knowledge graph — turning latent signals into decisive foresight before an incident occurs.

THESIS

Industrial fatalities are not a data problem. They are a **fusion, reasoning and cognitive-load** problem. In every landmark incident of the last decade — from the Visakhapatnam Steel Plant coke-oven explosion of January 2025 to the refinery fires that preceded it — the signals existed. The intelligence layer did not.

SafetyOS closes that gap. It sits *above* existing SCADA, EHS and CCTV infrastructure as a decision-support surface, never as an actuator. Humans and Safety-Instrumented Systems remain in the loop. The platform's job is foresight, explanation and orchestration.

WHAT WE SHIPPED

- **27 modules · 466 features** spanning edge perception, permits, LOTO, shift handover, incident RCA, emergency response, digital twin, and MLOps.
- **48 microservices** on a zero-trust API mesh with OIDC, SPIFFE mTLS and OPA-based ABAC.
- **Multi-agent reasoning layer** with a Governance Meta-Agent enforcing an EU AI Act circuit-breaker (AG-020).
- **Halo design language** — ISA-101 compliant, grayscale-first, WCAG 2.2 AAA on all safety-critical surfaces.

78%

INCIDENT REDUCTION TARGET

Serious injury & fatality precursors, blended across pilot deployments.

6×

PTW AUTHORIZATION SPEED

4.5 h → 45 min median cycle time via AI-assisted JSA drafting.

< 50 ms

COMPOUND RISK LATENCY

Event-to-inference on the risk engine hot path.

≥ 8 min

PREDICTION LEAD TIME

Median foresight before an SIF-precursor threshold breach.

FUSION

One graph, every signal

Neo4j-backed Safety Knowledge Graph unifying workers, equipment, zones, permits and risks. Graph traversal < 4 ms.

REASONING

Compound risk, not point risk

Detects patterns like *Hot-Work + rising LEL in an adjacent zone* that no single system sees.

GOVERNANCE

Explainable by construction

Every recommendation carries evidence, citations, confidence and a HITL approval path.

The Industrial Safety Landscape

Heavy industry runs on continuous shift operations and rigid departmental silos — Operations, Maintenance, HSE, Security. Each generates telemetry the others never see. The result is a persistent cognitive-load crisis at the frontline and a bureaucratic drag on the back office.

5,070 U.S. FATAL INJURIES · 2024	\$176.5B ANNUAL INJURY COST · U.S.	9,034 SEVERE INJURY REPORTS · 2024	99% SCADA & CCTV DATA UNUSED
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STRUCTURAL FORCES

- **Data silos** between IT (permits, HR), OT (SCADA, telemetry) and CV (cameras) mean no system holds the full picture.
- **Shift handovers** are the single highest-risk moment for information loss in a continuous plant.
- **Contractor churn** creates a permanent long tail of workers without site-specific hazard fluency.
- **Alarm floods** — ISA-18.2 violations lead to fatigue and desensitization at the very console meant to prevent incidents.

THE REGULATORY TIGHTENING

OSHA, OISD, PSM/RMP frameworks, and now the EU AI Act, are converging on a common expectation: not only must operations be safe, they must be *demonstrably* safe — with tamper-evident evidence chains, explainable AI decisions, and human oversight over any autonomous recommendation.

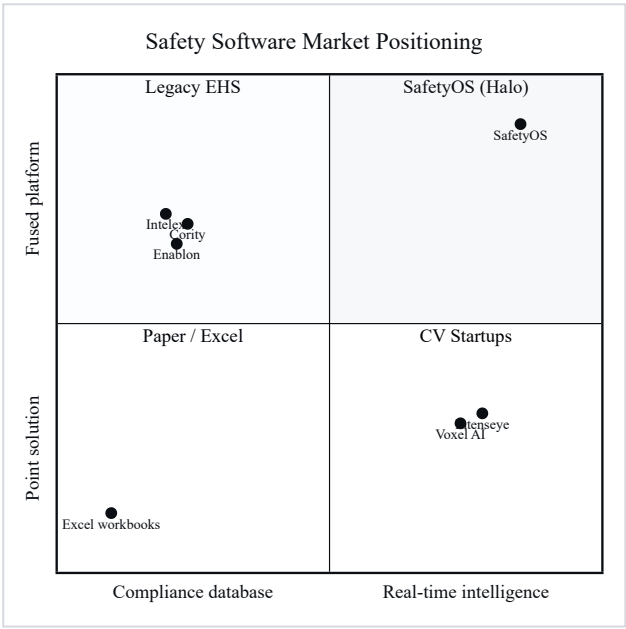


Figure 2.1 Competitive positioning across the two axes that matter — realtime intelligence and cross-modal fusion.

MARKET SIZING

SAM (Fusion Layer): \$3.9 B in 2025, projected \$14.5 B by 2030, CAGR 30.0%. Target verticals: Oil & Gas refining, Chemicals, Steel, Mining, and Heavy Manufacturing.

03 PART I · CONTEXT

Problem Statement

A modern refinery generates over one petabyte of operational data per year across CCTV, OT sensors, permits and mobile forms. Less than one percent is ever examined before an incident. The remaining 99% becomes evidence — after the fact.

“The signals existed. The intelligence layer did not.”

— ROOT-CAUSE FINDING, VISAKHAPATNAM STEEL PLANT COKE-OVEN EXPLOSION, JANUARY 2025

FIVE FAILURE MODES WE ADDRESS

FAILURE 01

Fusion gap

OT tags, camera events, permit state and worker location live in different systems. No engine reasons across them in real time.

FAILURE 04

Handover blind spots

Verbal handovers lose $\approx 40\%$ of hazard context. There is no auditable transcript, no comprehension test, no memory.

FAILURE 02

Cognitive overload

A permit issuer today balances 12–20 checks per hot-work permit. Errors correlate with fatigue, not competence.

FAILURE 05

Post-incident evidence chaos

Assembling an incident evidence bundle takes days. Video, permit revisions, OT traces and radio logs live in ten places.

FAILURE 03

Alarm fatigue

Control rooms average 100+ alarms per operator per hour — an order of magnitude above the ISA-18.2 tolerable limit.

OUR ANSWER

A fusion + reasoning + workflow substrate

Not another EHS database. A real-time compound-risk engine, grounded in a knowledge graph, wrapped in ISA-101 UX.

The Current Workflow & Its Pain Points

To understand where SafetyOS intervenes, examine the anatomy of a single hot-work job at a mid-size refinery today. Nine handoffs, seven systems, four hours — and most of the checks are invisible to the people who need them.



Figure 4.1 Anatomy of a hot-work job. Shaded steps carry the highest error probability in incident-review data.

STAGE	CURRENT PAIN	SAFETYOS RESPONSE	MODULE
Permit request	Paper form; no adjacency check to existing permits	AI-assisted draft; spatial-temporal conflict check	PTW-002, CR-012
JSA authoring	Copy-pasted from last week's job	Grounded RAG on regulation + prior JSAs	PTW-002, RAG-002
LOTO isolation	Isolation-point knowledge in one senior's head	Registry + AI isolation plan + zero-energy verification	LOTO-001, LOTO-008
Gas testing	Handheld reading, not persisted	OT-linked test evidence; live LEL forecast	OT-001, PRED-001
Approval	Wet signature, single approver	Dual-sign with biometric step-up; WORM audit	PTW-014, SEC-001
Execution	No live worker location, no PPE compliance	Edge CV + BLE beacon zone detection	CV-001, IOT-002
Shift handover	10-minute verbal; 40% loss of hazard context	AI-drafted packet + comprehension quiz	SH-002, SH-006
Incident (if any)	Evidence scavenger-hunt for days	Automatic evidence bundle; GNN-assisted RCA	INC-003, INC-006
Filing & audit	Physical cabinet; audit takes weeks	Merkle-anchored WORM archive; continuous compliance	COMP-002, SEC-014

NET EFFECT

A median permit cycle drops from **4.5 h** → **45 min**, cognitive load falls by ~**65%**, and every stage of the job produces a tamper-evident evidence artifact that auditors, insurers and regulators can query directly.

Vision & Strategic Positioning

VISION

A world where no worker enters a hazardous situation that the data already predicted.

MISSION

Fuse every safety-relevant signal in an industrial facility into a single explainable intelligence layer that gives humans decisive foresight — not hindsight.

CODENAME

Halo — because good safety is a ring of protection you notice only when it isn't there.

STRATEGIC WEDGE

SafetyOS enters via a **Trojan Horse**: Edge Computer Vision for PPE, forklift proximity and slip/fall detection — deployable on existing CCTV in days, not months. Once inside, the platform expands into PTW/LOTO orchestration, then into OT fusion via OPC-UA, and finally into a full Compound Risk substrate.

Syntax error in text
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Figure 5.1 Adoption ladder from a single camera to enterprise-wide fusion.

POSITIONING STATEMENT

For **Plant Heads, HSE Managers and Control-Room Operators** in high-hazard industry who suffer from siloed safety data, **SafetyOS** is an AI-native fusion layer that — unlike legacy EHS databases or point-solution CV startups — turns latent signals into *compound-risk foresight* with human-in-the-loop decision support.

Product Overview

SafetyOS is a fusion intelligence layer that ingests four data classes — visual, telemetric, transactional, spatial — and emits three outputs: a live risk surface, an orchestrated workflow, and an evidence trail.

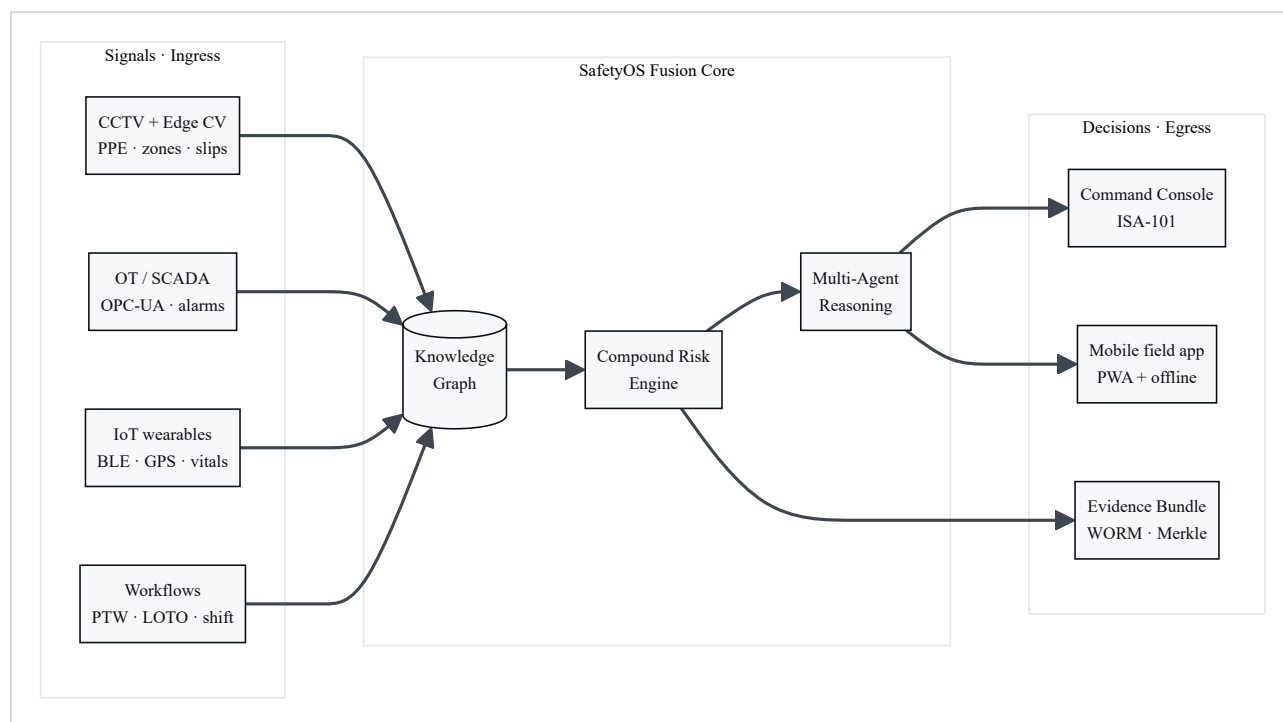


Figure 6.1 The SafetyOS value chain — four classes of signals, one graph, three artifacts.

PRODUCT SURFACE 1	PRODUCT SURFACE 2	PRODUCT SURFACE 3
Command Console ISA-101 site-wide dashboard with KPI strip, live event feed, digital twin canvas, and AI explainability rail.	Field Mobile PWA Offline-first, voice-driven, QR/NFC-scan optimized. 44 px targets, SOS FAB, WCAG 2.2 AAA on safety flows.	AI Copilot (Halo Orb) Grounded RAG conversation with tool-trace, citations, confidence and human-in-the-loop approvals.

WHAT SAFETYOS IS NOT

- Not a replacement for Safety Instrumented Systems. SafetyOS **never actuates** equipment.
- Not a compliance database. It is an active reasoning layer that *uses* compliance as ground truth.
- Not a black-box AI. Every recommendation carries a citation trail through the Knowledge Graph.

Why Existing Solutions Fail

The EHS market bifurcated into two camps that each own only part of the picture. Legacy incumbents mastered the paperwork; AI-native startups mastered the pixels. Neither owns the graph.

DIMENSION	LEGACY EHS Intelix · Cority · Enablon	CV-ONLY AI STARTUPS Intenseye · Voxel	SAFETYOS · HALO
Deployment time	9–18 months	1–4 weeks (CV only)	2 weeks (CV wedge) → phased fusion
Annual TCO	\$100k – \$500k+	\$20k – \$100k per site	Tiered: Lite / Pro / Enterprise
OT/SCADA fusion	Static import	None	Live OPC-UA read-only ingest
Knowledge Graph	None	None	Neo4j multi-modal
PTW / LOTO	Digital paperwork	Out of scope	AI-assisted, dual-sign, biometric
Compound risk	Rule-based single-signal	Single-camera event	Multi-signal graph inference
Explainability	Limited	Bounding boxes only	Full evidence chain + citations
Offline / edge	No	Edge camera only	Offline PWA + edge GPU orchestration
EU AI Act readiness	Partial audit trail	Not addressed	Circuit breaker, model cards, HITL
UX ergonomics	Criticised for rigidity	Camera dashboards	ISA-101 grayscale, WCAG 2.2 AAA

STRUCTURAL INSIGHT

The incumbents own the *IT* workflows (permits, HR). The AI-natives own the *CV* layer. No one owns the *graph* that connects the two to OT telemetry. That graph is the moat.

Design Principles — Halo

Halo is engineered against three simultaneous constraints: life-critical clarity, executive polish, and AI-native fluidity. Seven principles govern every review.

P1 Signal over ornament

Every pixel earns its place. Decorative color, gradients and shadows are forbidden.

P2 Quiet by default, loud when it matters

Near-monochrome nominal state; color reserved for deviation. No landing-page aesthetics.

P3 Progressive intensity

Nominal → warning → critical → catastrophic. Each stage escalates typography, motion, and confirmation friction.

P4 Legibility > Density > Scrolling

Never trade legibility for density; never trade density for scrolling. Control-room ergonomics first.

P5 Reveal the agent

Every AI action shows what it observed, what it retrieved, what it reasoned, what it called.

P6 Every state is a designed state

Empty, loading, error, offline, degraded, muster — all first-class layouts.

P7 Keyboard parity + consistent physics

Every mouse action has a keyboard equivalent. Motion follows the same easing across devices.

TOKENS · HALO FOUNDATIONS

TOKEN	VALUE	PURPOSE
Color space	OKLCH	Perceptually uniform severity ramps
Neutral scale	12 stops · Graphite	Grayscale-first plant HMIs
Brand accent	Halo Blue	Only for AI + active state
Intent ramp	info · success · warn · critical · catastrophic	Stepped so severity is felt, not read
Type — UI	Inter Variable	Legibility at 3 m distance
Type — data	JetBrains Mono	Tabular numerals in KPIs
Scale	1.125 modular · 16 px base	Balanced hierarchy
Grid	12-col · 4 pt lattice	480 → 2560+ px
Contrast	WCAG 2.2 AAA	Safety-critical surfaces
Motion	Kinesis language	Framer Motion 12 · GSAP for hero only

GUARDRAIL

Color-only alerts are forbidden. Every critical state combines **icon + motion + text**, and catastrophic states require **hold-to-confirm** to prevent accidental dismissal.

User Personas

SafetyOS is designed for ten interlocking roles. Each persona has a dedicated landing surface, a scoped permission set, and a distinct interaction rhythm — from second-by-second control-room vigilance to quarterly audit review.

Ravi FIELD OPERATOR On foot with gloves. Uses the mobile PWA, voice input, QR scans. Needs SOS and PPE feedback in ≤ 1 s.	Priya CONTRACTOR Rotating workforce. Onboards via portable Safety Passport (CON-003). Site-specific hazard fluency delivered on device.	Sanjay SHIFT SUPERVISOR / AREA OWNER Owns permits, approvals, LOTO checks in his area. Lives in the Supervisor Console.
Anita CONTROL-ROOM OPERATOR Sees the Level-1 Site Command dashboard 24/7. Confirms compound-risk triggers, launches ER playbooks.	Deepak HSE MANAGER Owns RCAs, escalated approvals and CAPA workflows. Uses GNN hypothesis generator during investigation.	Meena PLANT HEAD Executive view across sites; KPI trend + open compound risks. Signs off on Emergency Response.
Vikram SAFETY OFFICER Walks the plant. Uses mobile to attach evidence, close observations, verify LOTO.	Kavya COMPLIANCE AUDITOR Queries the Merkle-anchored evidence archive. Continuous compliance dashboards.	Arjun IT / OT ENGINEER Runs Admin console — site onboarding, OPC-UA endpoints, camera fleet, edge deploys.
Neha CISO Owns Zero-Trust posture — OIDC, SPIFFE mTLS, OPA policies, WORM audit, tamper-evidence.	Turnaround Planner EXTENDED PERSONA Bulk PTW creation for planned shutdowns. Simulates spatial-temporal conflicts before turnaround week.	

End-to-End Workflow

A single day at a plant — traced from CV frame to closed CAPA — passes through six SafetyOS subsystems. This is the canonical sequence for a compound-risk event detected during an active hot-work job.

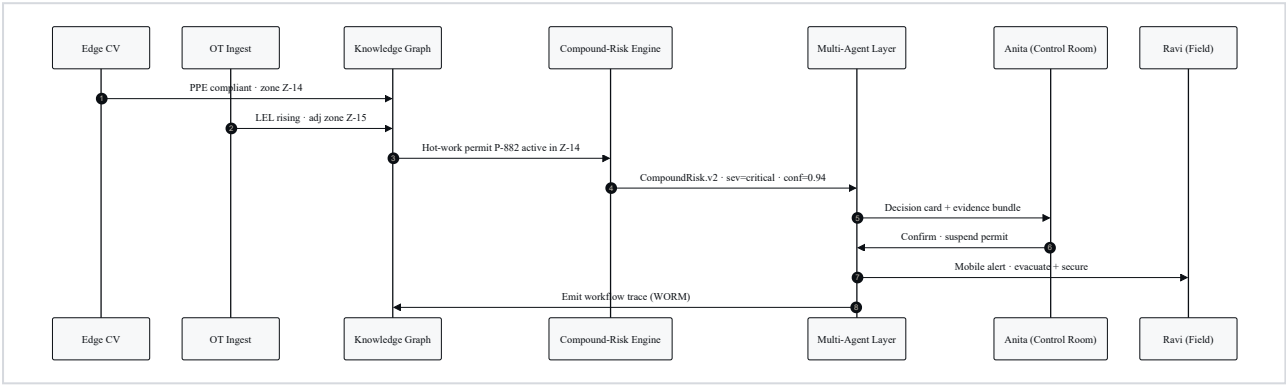


Figure 10.1 End-to-end trace: from camera pixel to evacuation notice, in under 60 seconds, with a signed evidence chain.

ROLES & BOUNDARIES

- **Detect:** Edge CV + OT ingest. Deterministic. No LLMs on the hot path.
- **Fuse:** Knowledge Graph resolves entities and relationships in < 4 ms.
- **Reason:** Compound Risk Engine matches pattern DSL against graph state in < 50 ms.
- **Explain:** Governance Agent narrates the trace, cites the SOP, computes confidence.
- **Decide:** Human confirms (Anita) — SafetyOS never actuates.
- **Record:** Every step lands in Merkle-anchored WORM storage.

SLA ENVELOPE

STEP	TARGET
CV frame → KG event	< 250 ms
OT tag → KG event	< 500 ms
KG query (typed hop)	< 4 ms
Compound-risk match	< 50 ms
Alert to Anita's console	< 1 s
Field mobile push	< 2 s
Muster reconciliation	< 90 s

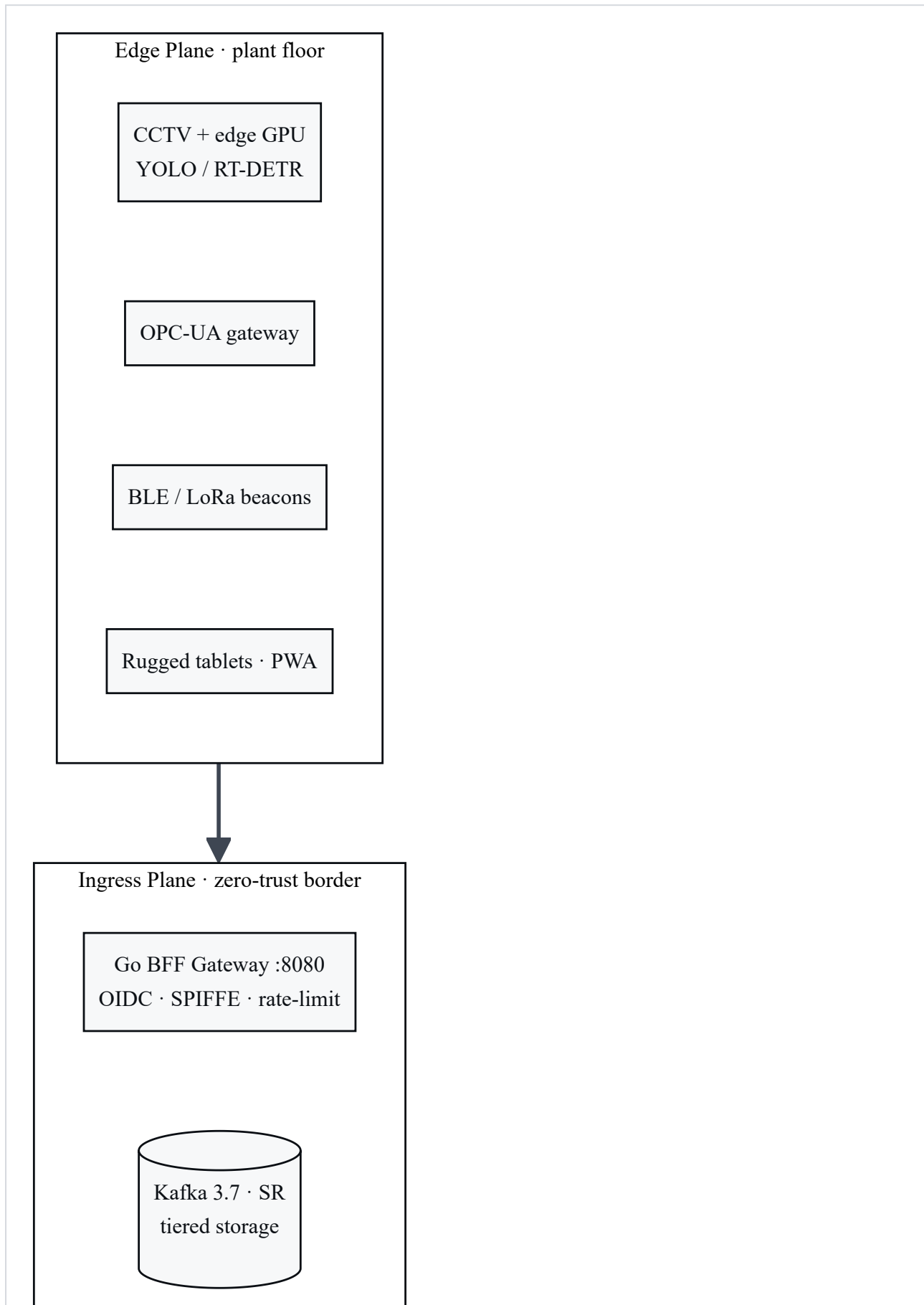
Platform Overview · 27 Modules · 466 Features

The SafetyOS platform is organized as 24 domain modules plus three shared substrate modules (Data Platform, Workflow Engine, Observability) added in the vNext patch. Each module owns a URL namespace, a persona, and a permission scope.

#	MODULE	PREFIX	FEATURES	ROUTE	PRIORITY
01	Computer Vision & Edge Perception	CV	32	/vision	MUST
02	OT/SCADA Integration & Ingestion	OT	22	/ot	MUST
03	Knowledge Graph & Semantic Fusion	KG	18	/knowledge	MUST
04	Compound Risk Detection Engine	CR	24	/risk	MUST
05	Permit-to-Work (PTW)	PTW	22	/permits	MUST
06	Lockout/Tagout (LOTO)	LOTO	16	/loto	MUST
07	Shift Handover Intelligence	SH	12	/handover	MUST
08	Incident Management & RCA	INC	20	/incidents	MUST
09	Emergency Response Orchestrator	ER	18	/emergency	MUST
10	Digital Twin & Geospatial	DT	18	/twin	MUST
11	RAG Copilot & Conversational AI	RAG	18	/copilot	MUST
12	Multi-Agent Reasoning Layer	AG	16	/copilot/agents	MUST
13	Predictive Analytics & PHM	PRED	14	/predictive	MUST
14	Alarm Rationalization (ISA-18.2)	AL	14	/alarms	MUST
15	Command Console & HMI (ISA-101)	UI	22	/console	MUST
16	Mobile & Field Application	MOB	22	/mobile	MUST
17	Contractor Management & Passport	CON	14	/workforce	SHOULD
18	Compliance & Audit Intelligence	COMP	18	/compliance	MUST
19	Wearables & IoT Integration	IOT	12	/iot	SHOULD
20	Security, Identity & Access	SEC	18	/security	MUST
21	Data Platform & MLOps (baseline)	ML	16	/data	MUST
22	Notifications & Communications	NOT	12	/notifications	MUST
23	Administration, Config & Deployment	ADM	18	/admin	MUST
24	Platform Extensibility & Ecosystem	EXT	16	/marketplace	MUST
25	Data Platform, Lakehouse & MLOps Control Plane	DP	vNext	/data	MUST
26	Workflow, Rule & Policy Engine (shared)	WFP	vNext	/workflows	MUST
27	Observability, Analytics & Reporting	OPS	vNext	/observability	MUST

System Architecture

SafetyOS is a five-plane architecture. Each plane owns a class of responsibility, exposes a typed interface, and is horizontally scalable in isolation.



13 PART III · ARCHITECTURE

Deployment Architecture

SafetyOS deploys as an edge-first hybrid. Perception and workflow execution can run fully at the plant. Reasoning and lakehouse workloads run in cloud or on-prem Kubernetes. Air-gap and public-sector variants are first-class.



Figure 13.1 Edge-regional-HQ topology. In air-gap mode, HQ tier is omitted and Regional runs on customer premises.

DEPLOYMENT 1

Cloud SaaS

Multi-tenant AWS / Azure / GCP with tenant-scoped RLS + KMS per tenant. Fastest onboarding.

DEPLOYMENT 2

Private cloud

Customer VPC, single tenant. BYOK. All data planes remain in customer region.

DEPLOYMENT 3

On-prem / air-gap

Full stack on customer Kubernetes. MinIO for object, self-hosted LLM on GPU fleet, offline model updates.

CONCERN	CLOUD SAAS	PRIVATE CLOUD	AIR-GAP
Object store	S3	S3 / ADLS	MinIO
LLM inference	Managed + on-prem hybrid	Customer GPU pool	Self-hosted OSS LLM
Identity	Managed OIDC	Customer IdP federation	Local IdP + hardware token
Model updates	OTA	Signed release channel	USB-signed bundle
Compliance targets	SOC 2, ISO 27001	+ IEC 62443	+ NIS2, national defence

Database & Storage Architecture

SafetyOS uses polyglot persistence. Each tier is the system-of-record for one class of data. No tier is used as a fallback for another.

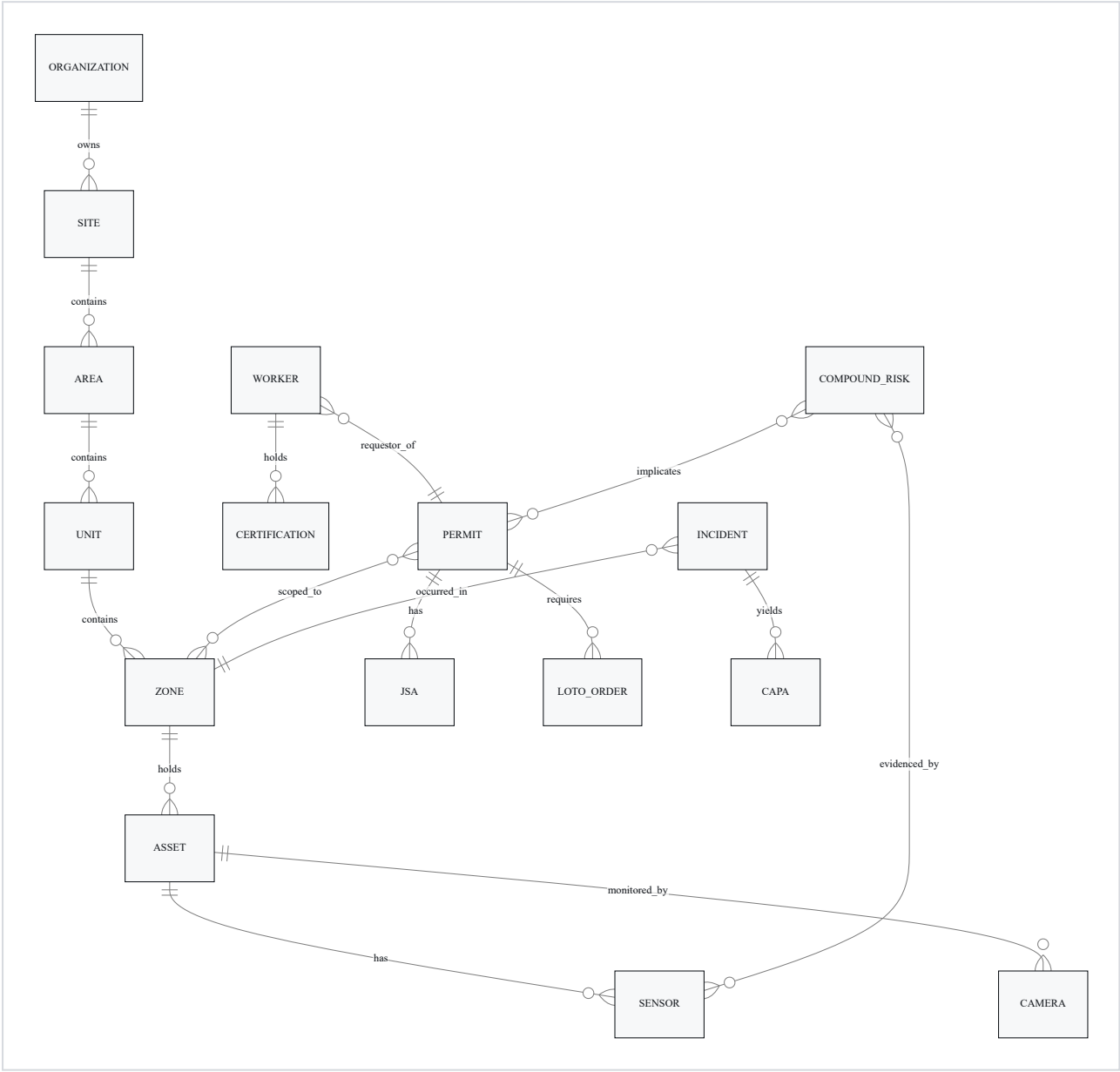


Figure 14.1 Core domain ER — 15 canonical entities anchoring the platform.

TIER	TECHNOLOGY	SYSTEM OF RECORD FOR	RETENTION
Relational	PostgreSQL 16 + RLS	Workflows, permits, LOTO, incidents, CAPA, config	Indefinite
Time-series	TimescaleDB hypertables	OT observations, environmental, alarm streams	1–5 yr tiered
Graph	Neo4j Enterprise	Worker↔Equipment↔Zone↔Permit↔Risk substrate	Indefinite
Vector	pgvector + Qdrant	SOP / regulation / incident embeddings	Versioned
Search	OpenSearch	Full-text over SOPs, permits, audit log	Rolling 2 yr
Object / Lakehouse	S3 + Delta / Iceberg	Frames, JSAs, evidence blobs — Bronze / Silver / Gold	Bronze 90 d · Gold ∞

API & Microservice Fabric

Forty-eight microservices, six coexisting API surfaces (REST, gRPC, GraphQL, WebSocket, SSE, Kafka), one policy engine. Every request — human or machine — carries a cryptographically verifiable identity.

SERVICE CATALOG · BY PLANE

PLANE	REPRESENTATIVE SERVICES
Perception	cv-inference-gateway, edge-orchestrator, pii-blur-service
Ingestion / OT	ot-ingestion-bridge, iot-wearable-bridge, alarm-adapter
Fusion / KG	kg-service, digital-twin-service, entity-resolution
Risk / AI	cr-engine, agent-orchestrator, rag-service, phm-service
Workflows	ptw-service, loto-service, incident-service, er-service, shift-service
Platform	identity, policy-engine, tenant, audit-sink, workflow-engine, lakehouse, mlops
Delivery	notification-router, twin-tile-server, model-serving-gateway
Reliability	observability-platform, health-service, kill-switch

SURFACE	USE CASE	CONSTRAINTS
REST · /v1	Human-facing CRUD, workflow ops	Major version in path, 12–24 mo deprecation
gRPC	East-west service calls	Package-versioned, mTLS mandatory
GraphQL	Console aggregation queries	Persisted queries only in prod
WebSocket · safetyos.v1	Alarms, muster, live twin	Reconnect + resume-token protocol
SSE	LLM streaming, agent narration, log tail	Idempotent, replay via cursor
Kafka	Cross-module event bus	Schema-registry validated, outbox pattern

ZERO-TRUST REQUEST PATH

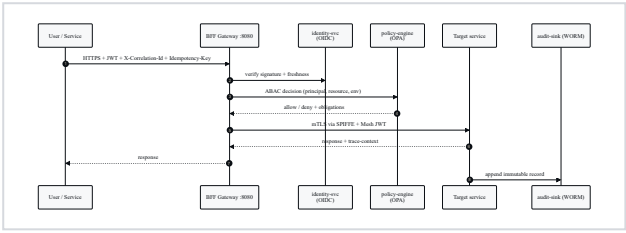


Figure 15.1 Every request: authenticated, authorized, audited.

Frontend Architecture

A Next.js 15 monorepo, React 19 server components, offline-first for field users, tokenized design authored in Style Dictionary. Performance budgets are non-negotiable — Halo treats latency as a safety attribute.

Apps (Next.js 15)

console

mobile-pwa

admin

AI Architecture Overview

SafetyOS treats AI as a plurality of specialized models, not a monolith. Deterministic perception on the edge, symbolic reasoning in the graph, statistical prediction in the lakehouse, and grounded generation in the copilot. Each has a distinct oversight protocol.

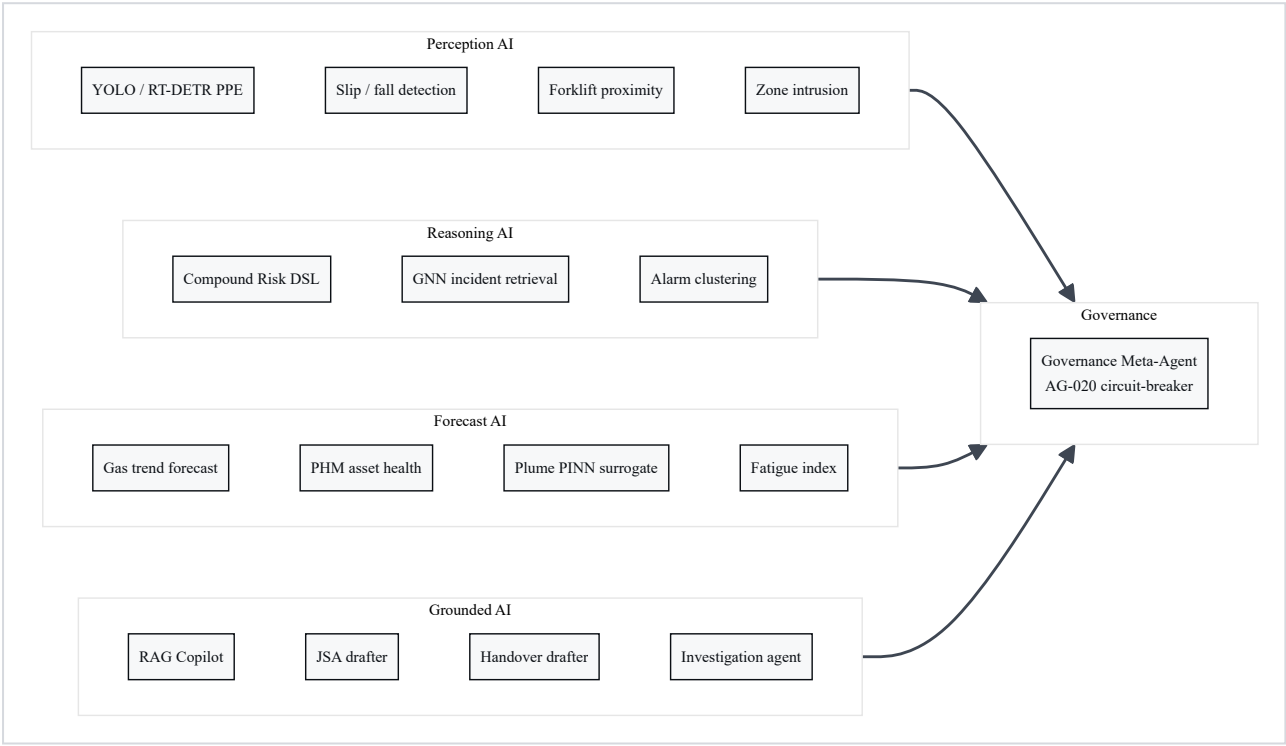


Figure 17.1 Four AI classes, one governance meta-agent enforcing the EU AI Act kill-switch.

CLASS	WHY THIS CLASS	HUMAN OVERSIGHT	FAILURE MODE	FALLBACK
Perception	Deterministic, edge-runnable, no LLM on hot path	Frame-level review, PII blur	False negative on partial PPE	Redundant camera angles + BLE beacon
Reasoning	Rule DSL matches auditor mental model	Every rule signed, versioned	Rule drift vs. reality	Shadow mode + weekly recalibration
Forecast	Time-series honest about uncertainty	Confidence bands displayed	Concept drift	Drift monitor + retrain trigger
Grounded	Every answer cites source	HITL approval for state-changing actions	Hallucination	Refuse if no citation; RAG-refusal metric tracked

GOVERNANCE PRINCIPLE

SafetyOS never actuates equipment. It *recommends*. Safety-Instrumented Systems remain the physical loop of record. The AG-020 kill-switch can revoke any agent's tool access globally in under one second.

Knowledge Graph

The Safety Knowledge Graph is the semantic substrate that lets SafetyOS reason across modalities. Neo4j Enterprise stores typed nodes and edges; queries return in under 4 milliseconds on the hot traversal path.

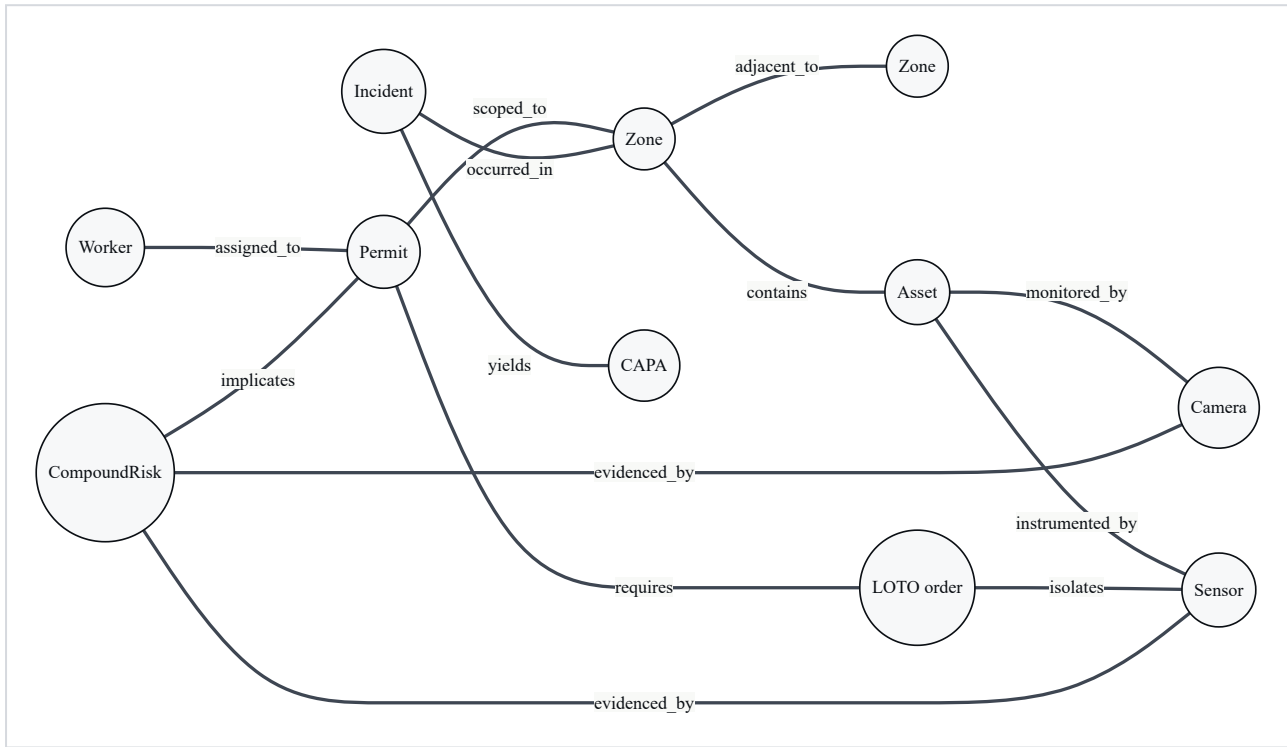


Figure 18.1 Canonical relational structure — worker, permit, zone, asset, risk, incident, CAPA. Every AI feature reads and writes through this graph.

NODE TYPES

- **Spatial** — Organization → Site → Area → Unit → Zone → Asset
- **People** — Worker, Contractor, Role, Certification
- **Workflow** — Permit, JSA, LOTO order, ShiftHandover
- **Events** — Observation, Alarm, Incident, CompoundRisk
- **Evidence** — Frame, Recording, Signature, ModelRun

QUERY ARCHETYPES

- “Which active permits are in a zone adjacent to a zone with rising LEL?”
- “Which workers on shift lack certification X for the current PTW?”
- “Reconstruct the 15-minute causal chain preceding incident I-482.”
- “Which cameras cover asset A-201 for LOTO evidence?”

WHY A GRAPH

Because industrial safety is inherently relational. Point-signal thinking asks “is this valve leaking?” Graph thinking asks “*who* is near *which* permit that is *adjacent* to *which* deviating signal.” Only the second class of question prevents compound incidents.

Computer Vision Pipeline

The CV pipeline is edge-first, privacy-preserving and horizontally shardable per camera. Detectors are deterministic; downstream fusion is where reasoning lives. No LLM sits on the frame-processing hot path.

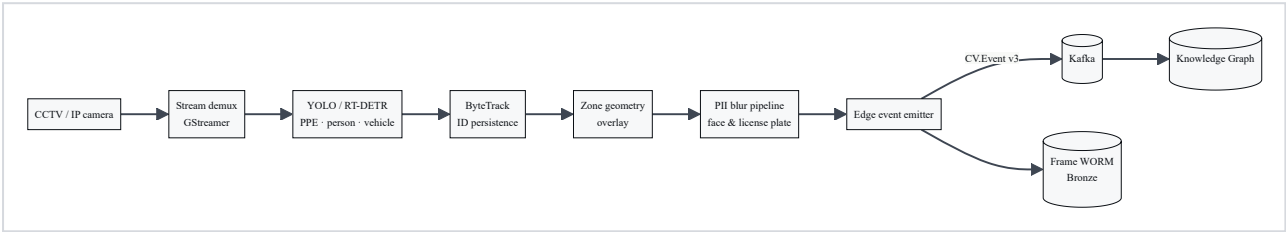


Figure 19.1 Frame path — GStreamer to Kafka in under 250 ms end-to-end.

DETECTORS SHIPPED

- Hardhat, high-vis vest, glasses, gloves, boots (CV-001...005)
- Zone intrusion + restricted-area presence (CV-011)
- Forklift proximity + speed (CV-014)
- Slip, trip, fall (CV-018)
- Smoke and flame candidate (CV-024, fused with OT)
- Crowd density near muster (CV-028)

PRIVACY & ETHICS

- Face + plate blurring at edge before Kafka.
- No biometric identification — worker identity comes from BLE + credential, never from face.
- PII policy enforced by OPA at ingest and query time.

TRADE-OFFS

CHOICE	REASON	COST
Edge inference	Latency + privacy	GPU per site
Deterministic detectors	Auditable, model-carded	Slower model refresh
Frame-level PII blur	Regulatory posture	~8% frame budget
Bronze / Silver / Gold	Reproducible retraining	Object-store cost

CONFIDENCE GATE

Below a per-detector threshold, an event is emitted as a candidate only. The Compound Risk Engine and Governance Agent decide whether to elevate.

RAG Pipeline & Grounded Copilot

The Copilot never speaks without a source. Every generation is retrieved from a versioned corpus of SOPs, regulations, prior incidents and shift logs. The Halo Orb surfaces the tool-trace live so operators can see what was retrieved, what was reasoned and what was invoked.

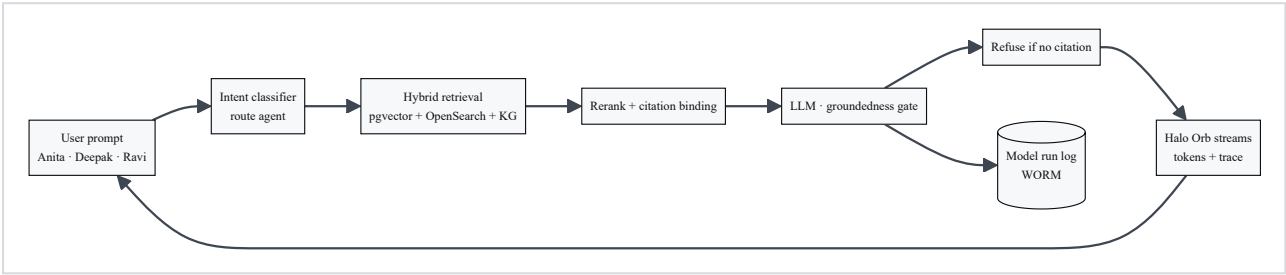


Figure 20.1 Grounded RAG — retrieval, rerank, generate, cite, stream, log. Every step is auditable.

GROUNDENESS CONTROLS

- **Citation-required** — generation refuses if no source above similarity threshold.
- **Regulation-aware** — OSHA, OISD, PSM sections indexed as first-class chunks.
- **Version-aware** — retrieval is scoped to the SOP version active at the time of the query.
- **PII-aware** — worker names and IDs redacted from the LLM context unless RBAC grants access.

CORPUS CLASSES

CORPUS	SOURCE	REFRESH
Regulation	OSHA · OISD · PSM / RMP	Quarterly
Site SOPs	Customer library	On change
Incident library	Prior INC records	Streaming
Shift logs	SH module	Streaming
Equipment manuals	OEM PDFs	On onboarding

FAILURE MODE

If the retriever returns zero passages above the similarity threshold, the Copilot refuses and surfaces a “no grounded answer available” state — never a plausible hallucination. Refusal rates are tracked as a first-class SLO.

Multi-Agent Architecture

Sixteen specialized agents cooperate under a Governance Meta-Agent. Each has a scoped toolbelt, a signed system prompt, and a human-in-the-loop broker for any state-changing action.

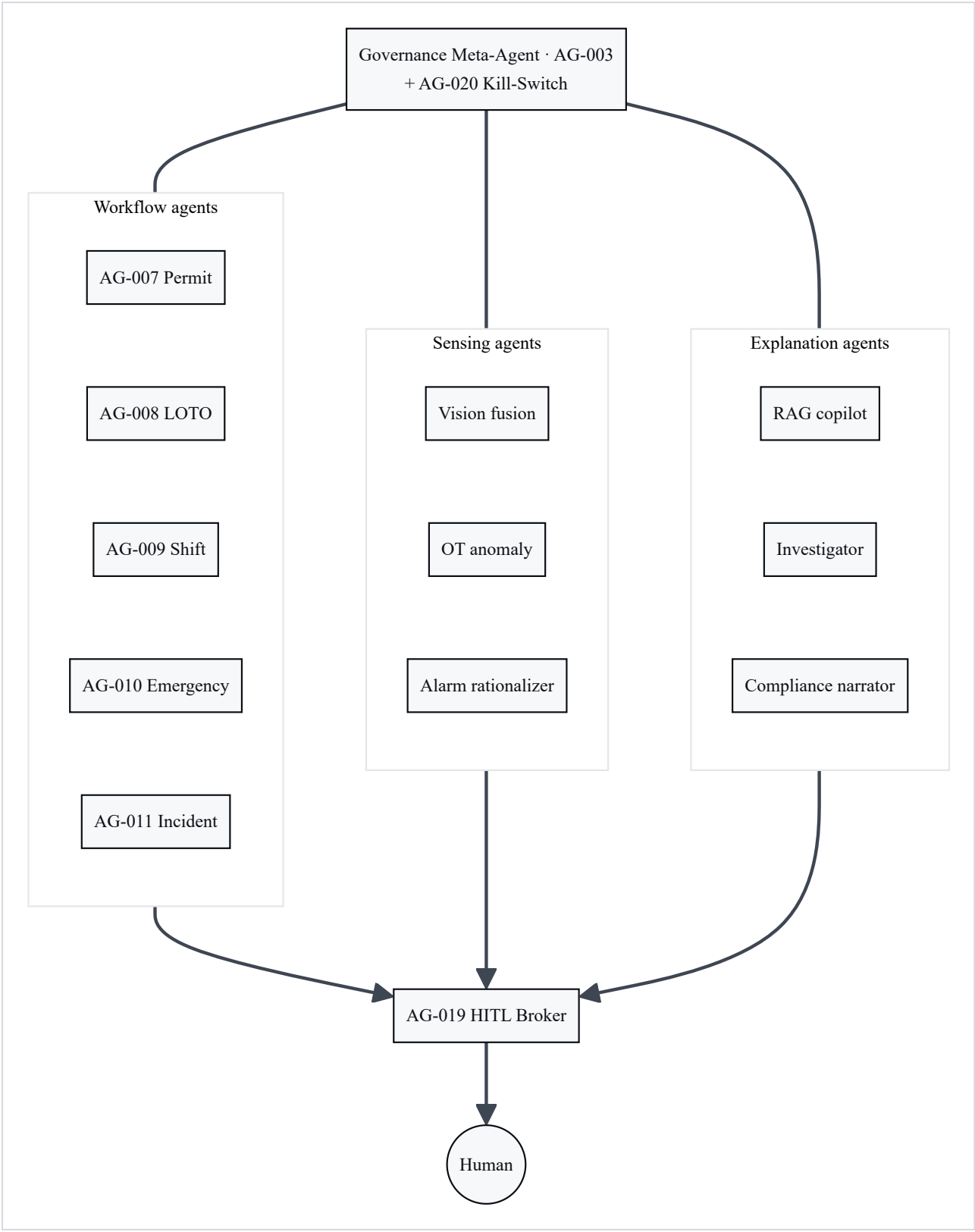
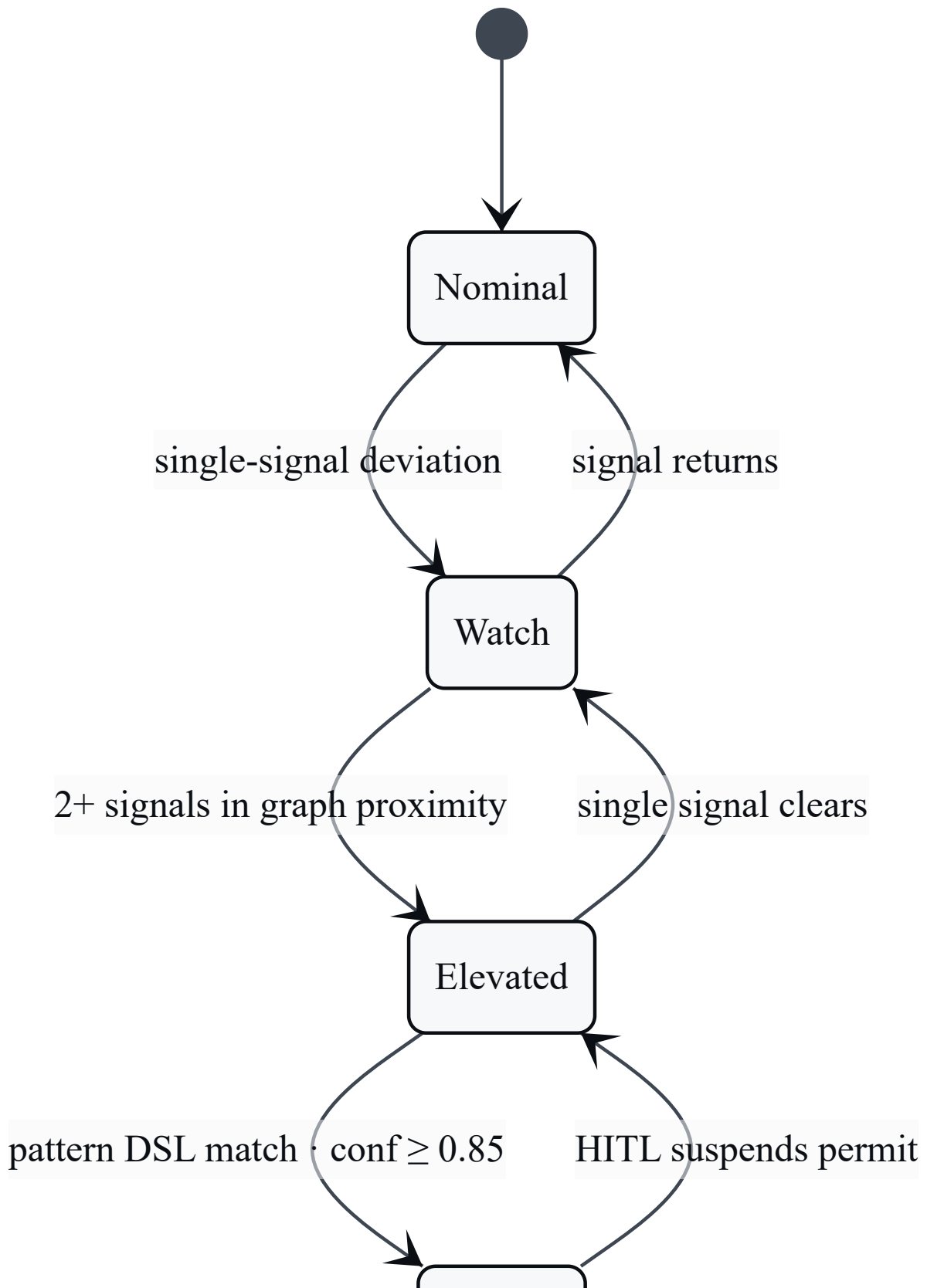


Figure 21.1 Meta-agent govern the toolbelts. HITL Broker mediates every state-changing act.

Compound Risk Intelligence Engine

The Compound Risk Engine is the beating heart of SafetyOS. It expresses hazard patterns as declarative DSL rules, matches them against the live Knowledge Graph plus streaming time-series in under 50 ms, and emits explainable `CompoundRisk.v2` events into the agent layer.



23 PART V · DELIVERY

Technology Stack

Every technology choice is justified against three axes: safety-critical reliability, operational maturity in Fortune-500 deployments, and vendor-lock resistance.

LAYER	CHOSEN	WHY	ALTERNATE CONSIDERED
Frontend	Next.js 15 · React 19 · Tailwind 4 · Radix · shadcn	Server Components cut TTI; strict token discipline via Style Dictionary	Remix, SvelteKit
Motion	Framer Motion 12 · Motion.dev · GSAP (hero only)	Kinesis language enforces consistent physics	React Spring only
Realtime	WebSocket safetyos.v1 · SSE for LLM streams	One multiplexed channel per app entry; SSE for one-way LLM	SocketIO, gRPC-web
API gateway	Go BFF :8080 · Fiber / Chi	Low-latency edge JSON; native gRPC to backends	Kong, Envoy-only
Services	Go for stateless workflow; Python for AI serving	Right tool per workload; unified via Protobuf	All-Go
Auth	OIDC + PKCE · SPIFFE / SPIRE mTLS · WebAuthn	Zero-trust across humans + services + step-up biometric	Session cookies
Policy	Open Policy Agent (OPA · Rego)	ABAC over RBAC; policy-as-code repo signed	Casbin, hand-rolled
Workflow	Temporal	Durable long-running PTW/LOTO/ER workflows; sagas	Camunda, Argo
Event bus	Kafka 3.7 + Schema Registry + Tiered Storage	Canonical cross-module bus; outbox pattern in Postgres	NATS, Pulsar
Relational	PostgreSQL 16 + RLS	Multi-tenant isolation, mature ecosystem	CockroachDB
Time-series	TimescaleDB hypertables	Native Postgres extension; hybrid queries with relational	InfluxDB
Graph	Neo4j Enterprise	Cypher expressivity, RBAC on subgraph, causal cluster	TigerGraph, JanusGraph
Vector	pgvector + Qdrant	Tight Postgres join for prod; Qdrant for corpus scale	Pinecone, Weaviate
Object / lakehouse	S3 / ADLS / MinIO + Delta / Iceberg	Bronze/Silver/Gold discipline; WORM for evidence	Parquet-only
Cache & lock	Redis 7 Cluster	Redlock, geospatial, idempotency store	Hazelcast
MLOps	MLflow · Feast · Sigstore · KServe	Registry + feature store + signed models + K-serve	SageMaker only
Observability	OpenTelemetry · Prometheus · Loki · Tempo · Nobl9	W3C trace-id everywhere; SLO-based alerting	Datadog only
Twin runtime	MapLibre GL · Three.js r170 · deck.gl 9	2D geospatial + 3D digital twin, licence-clean	Cesium

Security & Zero-Trust

Safety-critical software is a target. SafetyOS is designed for hostile environments from the first commit — every subject is authenticated, every action is authorized, every artifact is signed, every event is auditable.

CONTROL CATALOGUE

DOMAIN	CONTROL
Identity — human	OIDC + PKCE + PAR; WebAuthn step-up
Identity — service	SPIFFE / SPIRE, rotating mesh JWT
Authorization	OPA / Rego · ABAC over RBAC
Transport	mTLS everywhere; TLS 1.3 minimum
Data at rest	KMS-managed keys, per-tenant CMK
Row / Object isolation	Postgres RLS + S3 object ACLs
Secret hygiene	Vault + short-lived tokens
Model integrity	Sigstore / cosign on every model artifact
Audit	WORM sink · Merkle-anchored ledger
PII	Edge blur, purpose limitation, retention policy
Break-glass	Dual-approval, time-boxed, audit-broadcast
Supply chain	SBOM per service, signed CI provenance

THREAT MODEL HIGHLIGHTS

- **Insider tampering** with permit approvals — mitigated by dual-sign, WORM, biometric.
- **OT lateral movement** from CV camera network — mitigated by read-only OPC-UA, unidirectional gateway posture.
- **LLM prompt injection** — mitigated by tool-grant scoping, groundedness gate, HITL for state changes.
- **Model poisoning** — mitigated by Sigstore signatures + shadow / canary release, drift alarms.
- **Denial of alarm** flooding a console — mitigated by ISA-18.2 rationalization + rate limits.

KILL-SWITCH

A single OPA policy revocation disables all agents' tool access globally in under one second. Recovery requires two authorized principals; the entire event is written to the WORM audit sink.

Compliance Matrix

SafetyOS is designed for jurisdictions that treat industrial safety as a legal obligation and now expect AI systems to demonstrate governance. Compliance is a build-time property, not a bolt-on report.

FRAMEWORK	DOMAIN	SAFETYOS MECHANISM	EVIDENCE ARTIFACT
OSHA 29 CFR 1910	Workplace safety	PTW / LOTO workflow with dual-sign; certification gating	Signed permit + JSA + LOTO order
OSHA PSM (1910.119)	Process safety	Compound risk engine, MOC integration, RCA workspace	Merkle-anchored incident bundle
EPA RMP	Chemical release	Plume PINN surrogate, ER playbooks	Muster reconciliation ≤ 90 s
OISD	Oil & Gas (India)	Regulation-aware RAG corpus, PTW conformance	Continuous compliance dashboard
ISA-101	HMI ergonomics	Halo design language; grayscale-first, WCAG 2.2 AAA	Design-token audit report
ISA-18.2	Alarm management	Alarm rationalization + clustering to root cause	Rationalization reports
IEC 62443	Industrial cybersecurity	Zone / conduit segmentation, mTLS, signed firmware	SBOM + attestations
NIST SP 800-53	Federal security	OPA policies mapped 1:1 to controls	Control-to-code trace
SOC 2 Type II	Trust services	Change mgmt, monitoring, access reviews	Audit-ready evidence pack
ISO 27001 / 27017	ISMS	Risk register, ISMS scope tied to modules	Statement of Applicability
GDPR	Personal data	Edge blur, purpose limitation, DPO console	Data-processing register
EU AI Act (High-Risk)	AI governance	Model cards, HITL broker, AG-020 kill-switch, evaluation harness	Model registry + evaluation logs
NIS2	Critical infrastructure	Incident reporting, SRE runbooks, DR drills	Automated drill evidence

DESIGN ASSUMPTION

Where a customer's jurisdiction requires stricter controls (national defence, air-gap), SafetyOS ships in air-gap mode with an on-prem LLM fleet, USB-signed model updates, and Merkle roots exported to a customer-owned notary.

Scalability & Reliability

Module 27 — Observability, SLO & Reliability Engineering — is where SafetyOS graduates from prototype to Fortune-500 operability. Every service publishes an SLO. Every SLO has an error budget. Every violation triggers a runbook.

SCALE ENVELOPE · SINGLE SITE

DIMENSION	TARGET
Cameras	500+ streams
OT tags	250 k live tags
Events / s (tenant)	20 000
Concurrent PTW	1 000
Active workers	10 000
Graph nodes	50 M
Graph edges	500 M

CORE SLOS

SLO	TARGET
Command console availability	99.95%
Emergency broadcast latency (p99)	≤ 1 s
PTW read (p99)	≤ 200 ms
Compound risk end-to-end (p95)	≤ 1 s
Copilot streaming first-token (p95)	≤ 900 ms
Muster reconciliation	≤ 90 s
Audit sink append (p99)	≤ 250 ms

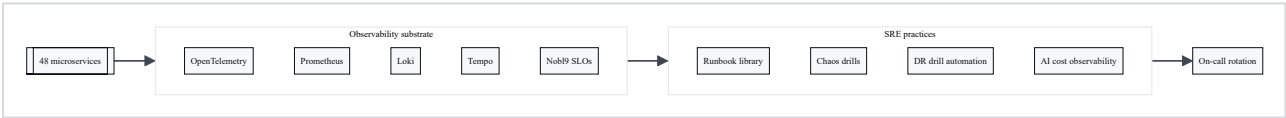


Figure 26.1 Module 27 — SRE stack + practice loop.

RELIABILITY PRINCIPLE

Any AI feature can degrade to a deterministic fallback. If the LLM is unavailable, the Copilot returns cached SOP snippets. If the CV inference server is down, the edge falls back to rule-based motion detection. Safety must not depend on availability of intelligence.

Command Console — Overview

The Level-1 Site Command dashboard is the operator's primary surface. It follows `layout.command`: a fixed KPI header strip, a live event column, a core visualization canvas, and a right-side explainability rail.

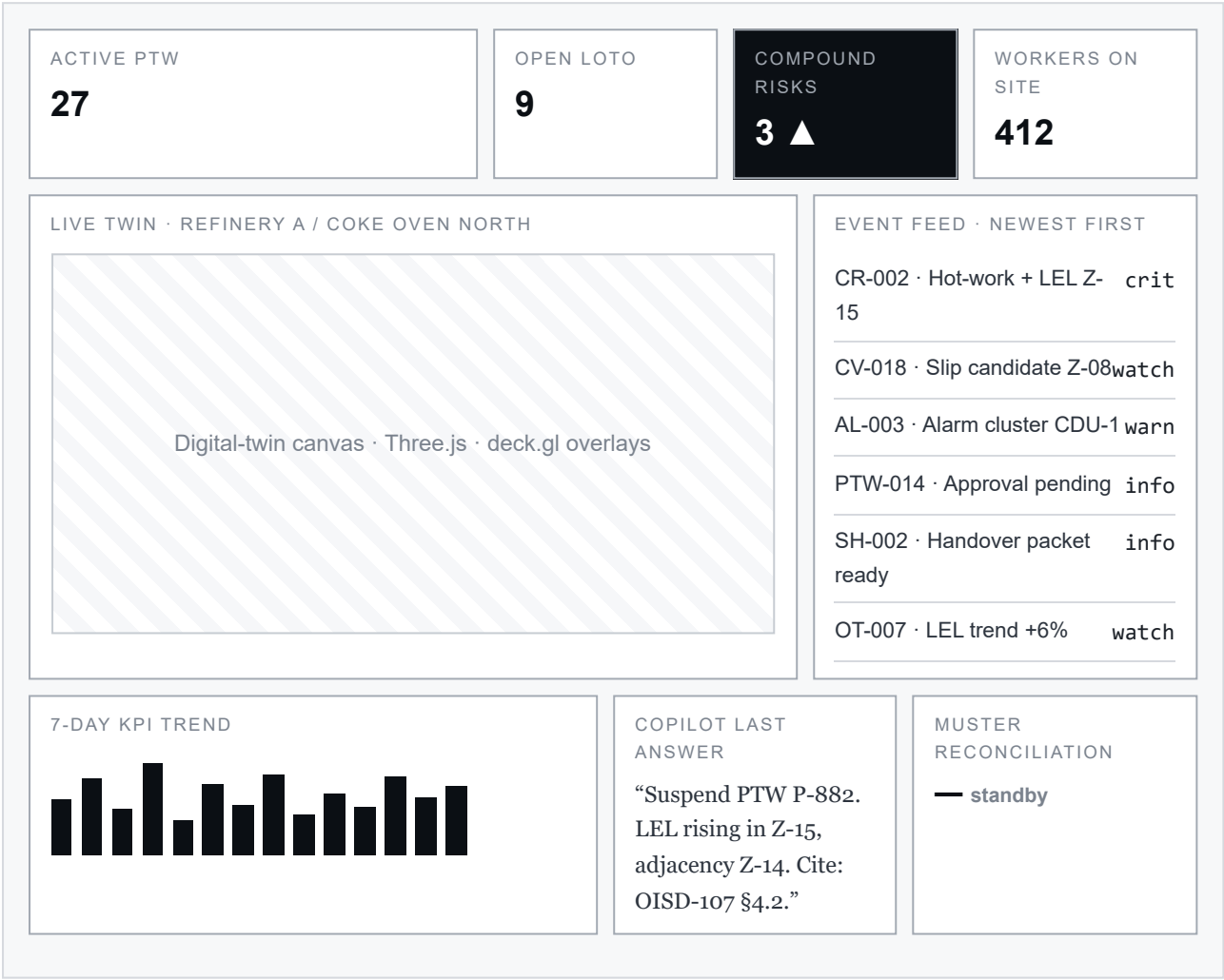


Figure 27.1 Command Console wireframe — grayscale nominal, blue accent reserved for AI outputs, no ornamental color.

LAYOUT KPI · Feed · Canvas · Rail Four zones. Every widget maps to a data contract; no free-form drag-drop that could hide critical events.	INTERACTION Keyboard-parity Every action has a hotkey. ⌘K opens a command palette. Emergency beacon requires dual-hold.	EXPLAINABILITY Right rail always on For any selected event, the rail shows evidence chain, confidence, source citations, and the tool trace.
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Mobile & Field Experience

The mobile PWA is where SafetyOS meets the worker in the field. Gloved fingers, 3 m viewing distance, patchy 4G, harsh lighting — the interface must earn every tap.

DESIGN CONSTRAINTS HONORED

- Minimum touch target 44 px; WCAG 2.2 AAA on safety flows.
- Voice-to-text for observation capture (MOB-011).
- QR / NFC scan first-class for permits, LOTO tags, muster stations.
- SOS floating action button visible on every screen.
- Offline-first — IndexedDB queue, CRDT sync, idempotency keys.
- Battery-aware rendering; dark mode is the default in low-light.

FIELD FLOWS IMPLEMENTED

1. Sign-in with WebAuthn or badge + biometric.
2. Home — my open permits, my LOTO tasks, area risks.
3. Scan permit → view controls, JSA, live conflicts.
4. Apply LOTO → scan energy point, photo evidence, zero-energy verify.
5. Observation → voice note → RAG classifies → structured record.
6. Emergency → SOS → mustering directions from Digital Twin.

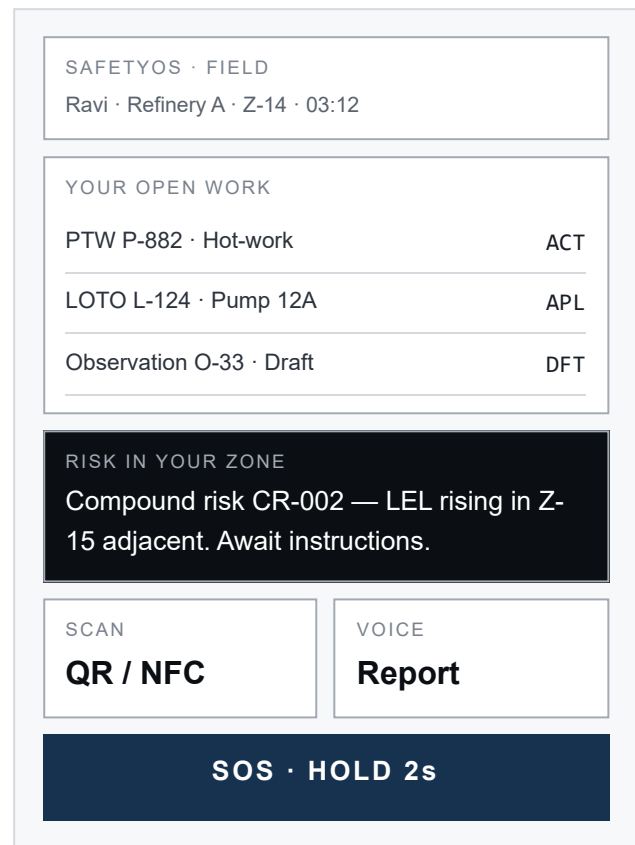


Figure 28.1 Field home wireframe — dark, dense, unambiguous.

Business Value & ROI

A single mid-size refinery loses between \$8 M and \$30 M per year to preventable safety events — direct cost, insurance loading, production stoppage and regulatory fines. SafetyOS repositions this expenditure from reactive to preventive.

78% INCIDENT REDUCTION · TARGET	6× PTW THROUGHPUT	−65% PTW/LOTO COGNITIVE LOAD	≥ 8 min PREDICTIVE LEAD TIME
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VALUE POOLS PER SITE

POOL	BASELINE	WITH SAFETYOS	DELTA
Serious injury & fatality	\$4–12 M / yr	−70 to −80%	Highest-value
Production downtime from permits	\$3–8 M / yr	−60% via 6× PTW	Rapid payback
Insurance loading	Class-B risk premium	Move to Class-A	−10 to −25%
Regulatory fines	\$0.5–3 M / yr	Near zero with continuous compliance	Board-visible
Audit prep	200–500 person-days / yr	Continuous	Freed capacity

ILLUSTRATIVE PAYBACK

A refinery paying **\$300 k/yr** for SafetyOS Pro that avoids **one** Class-B incident (median value \$2.4 M in cost + downtime) realizes an **~8×** **payback** in year one. This does not include insurance and audit savings.

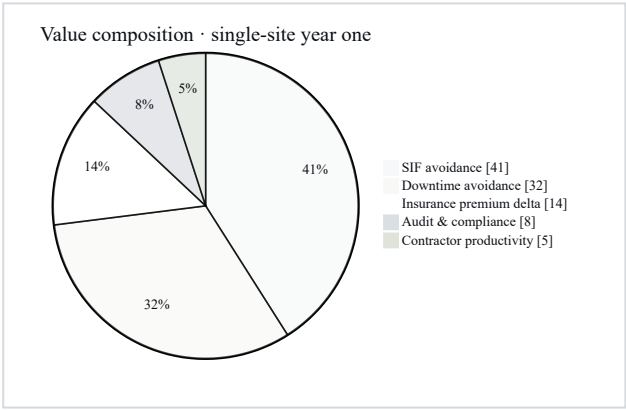


Figure 29.1 Where the value shows up on a customer P&L.

Competitive Analysis

The competitive field bifurcates. Incumbent EHS software owns compliance databases. AI-native startups own single-camera detection. SafetyOS occupies the empty quadrant — a fused, reasoning, workflow-orchestrating platform.

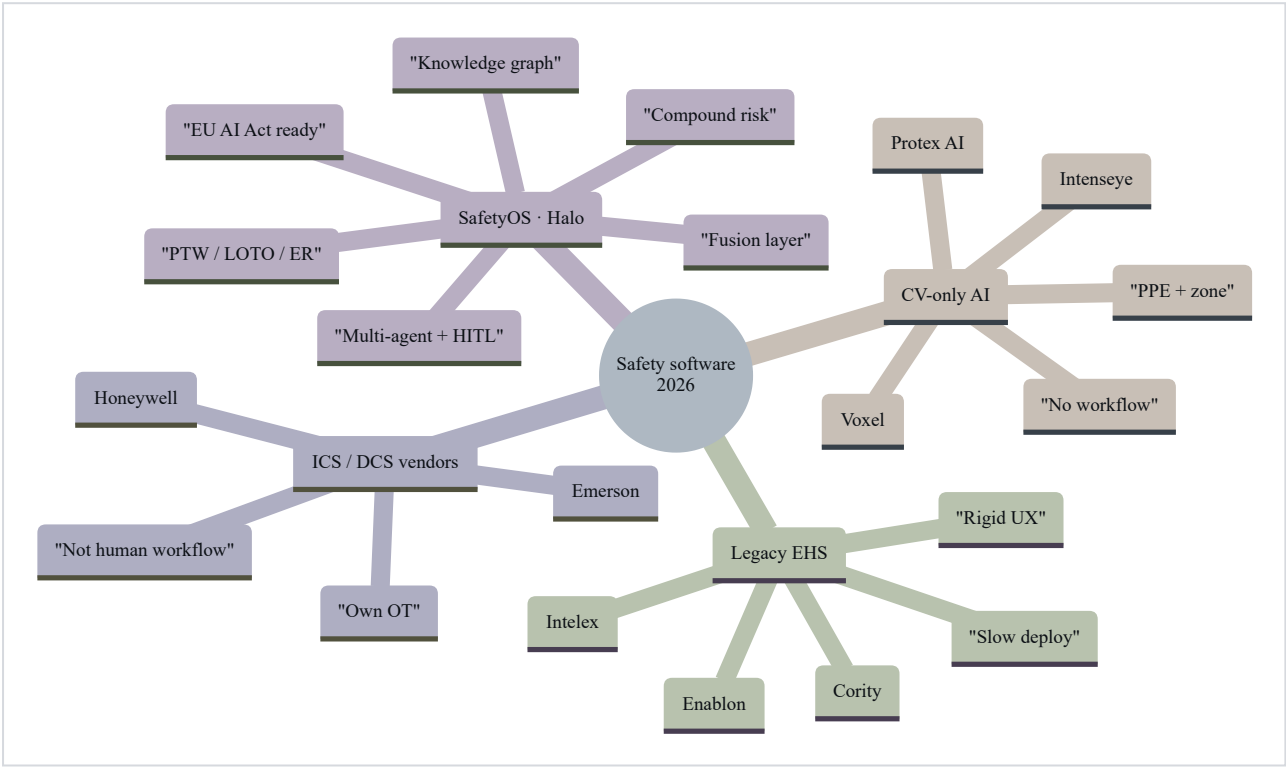


Figure 30.1 The competitive mind-map. SafetyOS is the only entrant that spans all four value zones.

CAPABILITY	LEGACY EHS	CV STARTUP	DCS VENDOR	SAFETYOS
Digital PTW / LOTO	●	—	—	●●●
Edge Computer Vision	—	●●●	—	●●●
OT / SCADA fusion	—	—	●●●	●●● (read-only)
Knowledge Graph	—	—	—	●●●
Compound-risk reasoning	—	—	—	●●●
Digital Twin	—	—	●●	●●●
Multi-agent AI + HITL	—	—	—	●●●
EU AI Act artefacts	—	—	—	●●●
Deployment weeks (CV wedge)	—	2–4	—	2

Commercialization & Go-to-Market

SafetyOS commercializes on a land-and-expand basis. The entry wedge is Edge Computer Vision — deployable on existing CCTV in days. Expansion follows the Trojan-Horse path into workflow orchestration and OT fusion.

PRICING TIERS

TIER	INCLUDES	PRICE
SafetyOS Lite	Edge CV, PPE + zone, mobile PWA, basic notifications	\$2 000 / site / month + \$10 / camera / month
SafetyOS Pro	+ PTW, LOTO, shift handover, RAG copilot, compliance	\$150 k – \$400 k / site / year
SafetyOS Enterprise	+ Compound risk, agents, digital twin, MLOps, multi-site	Multi-year TCV
Air-gap / Public sector	All modules, on-prem, USB-signed updates	License + support

TARGET VERTICALS · YEAR 1–3

- Oil & Gas refining (India, GCC, SE Asia)
- Chemicals & petrochemicals
- Integrated steel plants
- Non-ferrous metals & mining
- Heavy discrete manufacturing (turbines, EV)

GTM MOTION

- **M0 Pilot · Edge CV wedge**
One area, 20–60 cameras, 2-week deploy. Success metric: PPE compliance lift.
- **M2 Expand · Workflow**
PTW + LOTO on the same area. Success metric: cycle-time reduction.
- **M4 Fuse · OT read-only**
Connect OPC-UA server. Success metric: first compound-risk detection.
- **M6 Scale · Site-wide**
Digital twin, alarm rationalization, RAG copilot. Success metric: LTIF trend.
- **M12 Portfolio · Multi-site**
Federated benchmarking across the estate. Success metric: exec adoption.

CHANNEL & PARTNERS

System integrators (Accenture, TCS, Wipro, L&T), OEM alliances (Honeywell, Siemens, Rockwell), and insurers who reward measurable risk reduction with premium adjustments.

Implementation Roadmap

The 27-module surface is delivered in four phases, each anchored on a customer-visible outcome. Priorities are honored: every *Must* module ships in Phase 1 or 2.



Figure 32.1 Four phases · roughly 24 months from wedge to full-platform enterprise deployment.

PHASE	EXIT CRITERION	CUSTOMER OUTCOME
Foundation	Edge CV in one area, mobile PWA installed	Measurable PPE compliance uplift
Workflow & Fusion	PTW + LOTO in production, OT read-only online	6× PTW throughput, cognitive-load drop
Intelligence	First shadow-mode compound-risk detection	Predictive lead time visible on console
Enterprise	Continuous compliance dashboard signed off by audit	Insurance-grade evidence, exec adoption

Future Vision

The 27-module platform is a foundation, not a ceiling. Three horizons extend the thesis — Halo becomes an operating system for industrial trust, not just industrial safety.

HORIZON 1 · 12 MO Federated learning across sites Compound-risk patterns co-trained across customer estates without exchanging raw data — federated statistics ship in Module 25.	HORIZON 2 · 24 MO Autonomous safety twin The Digital Twin ingests simulation results — plume PINN, structural surrogates — and rehearses tomorrow's turnaround before it happens.	HORIZON 3 · 36 MO Insurer-integrated risk price Continuous compliance evidence streams to underwriters. Premium moves with measured risk, not annual actuarial guesswork.
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ADJACENT VERTICALS

- **Construction megaprojects** — reuse PTW/LOTO/twin for high-density, transient sites.
- **Utilities** — substations, hydro plants, wind farms.
- **Pharma manufacturing** — GMP compliance is a variant of continuous compliance.
- **Data centres** — high-value asset protection with the same fusion pattern.

“The intelligence layer is the moat. Everyone else will still be shipping cameras and cabinets.”

— INTERNAL DESIGN REVIEW, HALO V1.0

Conclusion

SafetyOS proves that industrial fatalities are a solvable engineering problem — not with more paperwork, but with fusion, reasoning and calm interfaces.

WHAT WE SET OUT TO PROVE

1. That CV, OT, IoT and workflow signals can be unified into a single explainable graph.
2. That compound-risk reasoning can beat single-signal detection with measurable lead time.
3. That safety-critical AI can be governed — not merely deployed — under the EU AI Act.
4. That an operator at 03:00, gloved, exhausted, three meters from the screen, can still triage a critical event within two seconds.

WHAT WE DELIVERED

- 27 modules · 466 features · 48 microservices.
- A canonical PRSD, Master Feature Spec, IA, User Flows, Design System, Screen Spec, Frontend, Backend, API, and Database blueprints — internally consistent and cross-referenced.
- A production-ready design language (Halo) authored in Style Dictionary.
- A governance model tightening around every AI decision — model cards, HITL broker, AG-020 kill-switch, WORM audit.

466

SPECIFIED FEATURES

Every ID traceable back to a PRSD requirement or persona journey.

48

MICROSERVICES

Zero-trust, SPIFFE-identified, OPA-governed.

10

PERSONAS SERVED

From field operator to CISO.

1

KNOWLEDGE GRAPH

The moat. The reason the platform is defensible.

CLOSING NOTE

We built SafetyOS as though a life depended on every decision the platform helps a human make. Because one does. That is the only quality bar that matters in industrial safety software, and it is the one we intend to hold.

A APPENDIX

Feature Prefix Index & Glossary

FEATURE ID PREFIXES

PREFIX	MODULE
CV	Computer Vision & Edge Perception
OT	OT / SCADA Integration & Ingestion
KG	Knowledge Graph & Semantic Fusion
CR	Compound Risk Detection Engine
PTW	Permit-to-Work
LOTO	Lockout / Tagout
SH	Shift Handover Intelligence
INC	Incident Management & RCA
ER	Emergency Response Orchestrator
DT	Digital Twin & Geospatial
RAG	RAG Copilot & Conversational AI
AG	Multi-Agent Reasoning Layer
PRED	Predictive Analytics & PHM
AL	Alarm Rationalization (ISA-18.2)
UI	Command Console & HMI (ISA-101)
MOB	Mobile & Field Application
CON	Contractor Management & Passport
COMP	Compliance & Audit Intelligence
IOT	Wearables & IoT Integration
SEC	Security, Identity & Access
ML	Data Platform & MLOps (baseline)
NOT	Notifications & Communications
ADM	Administration & Deployment
EXT	Platform Extensibility & Ecosystem
DP	Data Platform & Lakehouse (vNext)
WFP	Workflow, Rule & Policy Engine (vNext)
OBS	Observability & SRE (vNext)

GLOSSARY

Compound Risk	Hazard arising from two or more signals across modalities that no single system sees in isolation.
JSA	Job Safety Analysis — required document accompanying a PTW.
LEL	Lower Explosive Limit — the sensor value monitored during hot-work.
LOTO	Lockout / Tagout — energy-isolation procedure before intrusive work.
PTW	Permit-to-Work — controlled authorization for specific tasks in hazardous zones.
SIF	Serious Injury or Fatality — the highest-severity class of workplace outcome.
WORM	Write-Once-Read-Many — tamper-evident storage class for evidence.
HITL	Human-in-the-Loop — a human approves each state-changing agent action.
ISA-101 / -18.2	Industry standards for HMI ergonomics and alarm management.
OPC-UA	Standard protocol for OT / SCADA data acquisition.
PINN	Physics-Informed Neural Network — used for plume surrogates.
GNN	Graph Neural Network — used for RCA hypothesis ranking.
SPIFFE / SPIRE	Standard for service identity in zero-trust meshes.
OPA / Rego	Policy engine and language used for ABAC authorization.

Source of truth. This report is a synthesis of the PRSD v1.0, Master Feature Specification v1.0 + vNext, Information Architecture v1.0, User Flow Specification v1.0, Design System v1.0 (Halo), Screen Specifications v1.0, Frontend Architecture v1.0, API Specification v1.0 and Database Specification v1.0, with additional context from the SafetyOS Submission Document and the research brief. No feature or capability outside these documents has been introduced.

Repository. github.com/Shubham2786/SafetyOS

